B. The Third Side

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: geometry, greedy, math | https://codeforces.com/problemset/problem/2074/B

The pink soldiers have given you a sequence a consisting of n positive integers.

You must repeatedly perform the following operation **until there is only 1 element left**.

- Choose two **distinct** indices i and j.
- Then, choose a positive integer value x such that there exists a **non-degenerate triangle**[1] with side lengths a_i, a_j, and x.
- Finally, remove two elements a_i and a_j, and append x to the end of a.

Please find the maximum possible value of the only last element in the sequence a.

[1]A triangle with side lengths a, b, c is non-degenerate when $a+b$ gt c, $a+c$ gt b, $b+c$ gt a.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$).

The second line of each test case contains n integers $a_1,a_2,\ldots,a_n$ ($1 \le a_i \le 1000$) — the elements of the sequence a.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the maximum possible value of the only last element on a separate line.

Example

Input
4 1 10 3 998 244 353 5 1 2 3 4 5 9 9 9 8 2 4 4 3 5 3

Output
10 1593 11 39

Note

On the first test case, there is already only one element. The value of the only last element is 10.

On the second test case, a is initially [998,244,353]. The following series of operations is valid:

1. Erase a_2=244 and a_3=353, and append 596 to the end of a. a is now [998,596].
2. Erase a_1=998 and a_2=596, and append 1593 to the end of a. a is now [1593].

It can be shown that the only last element cannot be greater than 1593. Therefore, the answer is 1593.